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HR EXCELLENCE IN RESEARCH

05: Summary and Integrated Case Study

Network Applications

LM in Ingegneria Informatica

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Summary and Integrated Case Study

This final module serves as the synthesis of the entire curriculum, transitioning from theoretical components to a unified, real-world application. We will bridge the gap between network infrastructure, distributed intelligence, and high-level autonomous orchestration.

Core Objectives

- **Final Synthesis:** Connecting the dots from physical layers to the intelligence plane.
- **The "Full Stack" of 6G:** Demonstrating how hardware, software, and AI converge.
- **Real-World Application:** Exploring a multi-disciplinary Integrated Case Study on future transportation.

Orchestrating the 6G Pillars

The Recap: Our Building Blocks

- **The "Pipe" (SDN/NFV):** Flexible, software-defined conduits for data.
- **The "Brain" (DML/AI):** Distributed intelligence that processes complexity.
- **The "Edge" (MEC):** High-speed localized compute to bypass core latency.
- **The "Language" (ZSM/Intent):** The autonomous logic that translates goals into actions.

The Challenge: The 10ms Window

The transition from theory to practice is defined by **synchronization**. In a safety-critical Intelligent Transportation System (ITS), the most safety-critical loop may have only **a few to tens of milliseconds** to:

1. **Sense:** Gather sensor data via the *Pipe*.
2. **Process:** Analyze the threat via the *Edge* and *Brain*.
3. **Command:** Issue an autonomous *Intent* to the vehicle.

Defining the Case Study: Intelligent Transportation Systems (ITS)

ITS represents a multi-disciplinary challenge where 6G acts as the unifying fabric. It is defined as a "**System of Systems**" that orchestrates interactions between vehicles, smart infrastructure, and human actors.

The V2X (Vehicle-to-Everything) Ecosystem

To achieve autonomous safety, 6G must handle four primary communication vectors:

- **V2V (Vehicle-to-Vehicle):** Direct low-latency links between cars.
 - *Application:* Platooning (high-speed coordinated driving).
- **V2I (Vehicle-to-Infrastructure):** Communication with roadside units (RSUs).
 - *Application:* Traffic light synchronization and intersection management.
- **V2P (Vehicle-to-Pedestrian):** Connecting to smartphones/wearables.
 - *Application:* Proactive safety alerts for pedestrians and cyclists.
- **V2N (Vehicle-to-Network):** Cellular links to the core and cloud.
 - *Application:* Traffic-aware routing, cloud-assisted map updates, and fleet-level coordination.

Mission-Critical Requirements (The SLA)

To transform Intelligent Transportation from a vision to a reality, future 6G/IMT-2030 systems are expected to support very demanding design targets:

1. Extreme Reliability & Latency

- **Target:** Latency <5ms with 99.9999% (6 nines) reliability.
- **Purpose:** Instantaneous braking and collision avoidance where milliseconds represent the difference between safety and impact.

2. Ultra-High Throughput

- **Target:** Hundreds of Mbps to Gbps-class connectivity for selected vehicles or infrastructure nodes.
- **Purpose:** Supporting raw 8K video feeds from vehicle cameras and LIDAR sensor fusion sharing for collective perception.

3. Massive Connection Density

- **Target:** Supporting dense urban scale / up to 10^6 devices/km² benchmark.
- **Purpose:** Managing dense urban intersections where vehicles, roadside sensors, and thousands of pedestrian wearables compete for spectrum.

4. Robust Mobility

- **Target:** Seamless connectivity at speeds up to 250 km/h.
- **Purpose:** Ensuring safety and infotainment continuity for high-speed commuter trains and highway autonomous convoys.

The ITS Ecosystem Architecture

The Intelligent Transportation System (ITS) ecosystem is built upon a multi-layered architecture designed to enable ultra-reliable, low-latency communication (URLLC). By distributing intelligence from the vehicle to the cloud, the system ensures real-time safety and long-term efficiency.

1. On-Board Units (OBU)

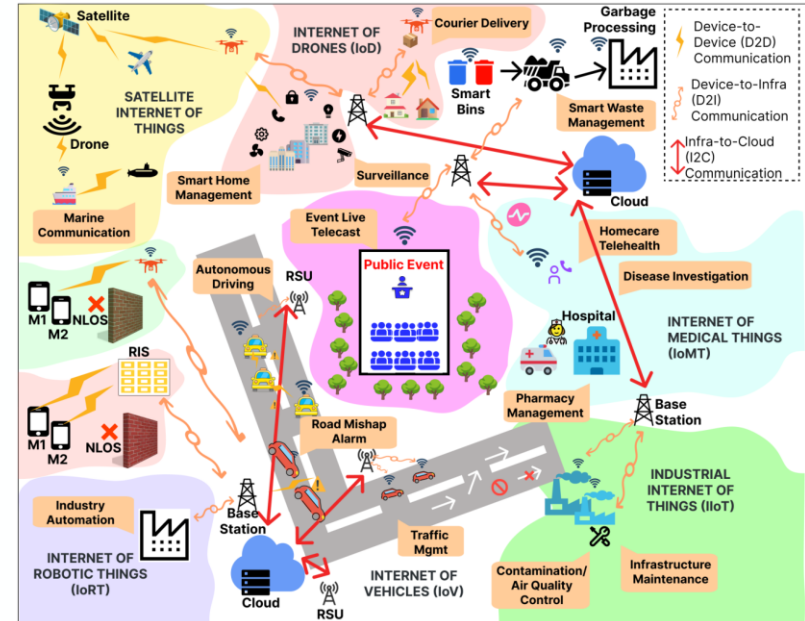
- Edge nodes within vehicles processing high-bandwidth data from **Lidar, Radar, and 4K Cameras** for local perception.

2. Road Side Units (RSU)

- Strategic **6G/next-generation base stations** with integrated Multi-access Edge Computing (MEC) for millisecond-level traffic coordination.

3. Central ITS Cloud

- Aggregated long-term repository for **Regional Traffic Management**, policy orchestration, and massive data analytics.



Step 1: Network Slicing for ITS Isolation

Architecting a 6G Intelligent Transportation System (ITS) requires that critical data types do not compete for resources. We move beyond "best effort" delivery to **logical isolation and resource guarantees** to ensure safety-critical commands are never delayed by non-essential traffic.

The Three Functional Slices

1. URLLC Slice (The Safety Plane)

- **Purpose:** Reserved for autonomous vehicle control, collision avoidance, and V2V safety heartbeats.
- **KPIs:** Sub-5ms latency and 99.9999% reliability.
- **Value:** Provides a protected environment for the highly reliable, bounded-latency delivery of emergency commands.

2. eMBB Slice (The Data Plane)

- **Purpose:** High-throughput delivery of Dynamic HD Maps, 4K/8K sensor sharing (Collective Perception), and infotainment.
- **KPIs:** Gigabit-per-second peak data rates.
- **Value:** Manages massive data volumes without impacting timing-sensitive safety signals.

3. mMTC Slice (The Telemetry Plane)

- **Purpose:** Connectivity for high-density, low-power urban sensors.
- **Applications:** Monitoring pavement temperature, air quality, and infrastructure health.
- **KPIs:** Supports 1 million+ devices per km² with high energy efficiency.

Architectural Value: By utilizing SDN/NFV to slice physical hardware, 6G creates specialized, isolated networks tailored to the distinct needs of the autonomous ecosystem.

Step 2: MEC for Distributed Decision Making

While 6G provides the "Pipe," **Multi-access Edge Computing (MEC)** provides the localized intelligence required for real-time autonomous safety. By moving compute power to the Road Side Unit (RSU), the system can avoid the tens-to-hundreds of milliseconds often associated with distant cloud processing.

Local Perception Fusion

The MEC server acts as a centralized orchestrator **for localized traffic**. It fuses high-fidelity data (video, LiDAR, Radar) from multiple cars and infrastructure sensors at a specific intersection to build a **local digital twin of the intersection**.

Solving the "Ghost" Hazard

- **Problem:** Individual vehicle sensors are limited by line-of-sight. A speeding bicycle or pedestrian might be hidden behind an obstacle from every vehicle's perspective.
- **Solution:** The RSU, positioned at a high vantage point or aggregating data from multiple angles, detects the hazard.
- **Action:** The MEC broadcasts an immediate "Stop" or "Avoid" command to all relevant vehicles within a <5ms window.

Step 3: SDN/NFV for Service Continuity

As vehicles move at high speeds (100 km/h or more), the 6G network must ensure that the intelligence supporting the vehicle is never left behind. This is achieved through the orchestration of SDN and NFV.

The Role of SDN (Software-Defined Networking)

- **Dynamic Re-routing:** As the vehicle moves, the SDN controller updates forwarding paths in the transport and edge network as the active serving point changes.
- **Path Optimization:** It ensures the shortest possible path between the On-Board Unit (OBU) and the active Road Side Unit (RSU), minimizing jitter and latency.

The Role of NFV (Network Function Virtualization)

- **VNF Migration:** The safety-critical applications are hosted as Virtualized Network Functions (VNFs) on the MEC.
- **Stateful Continuity:** As the car crosses into a new sector, NFV supports proactive state transfer or replication of the vehicle context to the next MEC server.

Step 4: DML for Privacy-Preserving Intelligence

How can autonomous vehicles navigate complex edge cases (e.g., dense fog, black ice) without compromising privacy by sharing raw dashcam footage or sensitive location data?

The Solution: Federated Learning (FL)

Distributed Machine Learning (DML) decentralizes intelligence by moving the model to the data rather than moving data to the model.

1. Local Training

Vehicles act as edge learners, training a **Local Model** on internal sensors when encountering new environments.

- **Privacy:** Raw video and GPS logs remain on the vehicle's local hardware.

Step 4: DML for Privacy-Preserving Intelligence

2. Weight Aggregation

The vehicle extracts mathematical "knowledge" (weights) and transmits these small updates to a central server or Roadside Unit (RSU) via 6G.

3. Global Model Update

The server aggregates updates from thousands of vehicles to refine a **Global Driving Model**.

- **Result:** Fleet-wide intelligence improves through collective experience.
- **Cycle:** The updated global model is pushed back to all vehicles, ensuring every car learns from events it never personally encountered.

Key Takeaway: 6G and edge infrastructure can support frequent, scalable, and privacy-preserving exchange of model updates, without requiring raw sensor data to leave the vehicle.

Step 5: Zero-Touch Service Management (ZSM)

In the final step of our integrated case study, we see how the 6G network manages itself through **Zero-Touch Service Management (ZSM)**.

The Challenge: Massive Event Spikes

- **Scenario:** A major stadium event concludes.
- **Impact:** 20,000 autonomous vehicles and 50,000 smartphones enter the network cell simultaneously.
- **Problem:** Traditional networks would suffer from "Signaling Storms" or congestion, delaying critical safety packets.

The ZSM Automation Logic

- **Observability:** The ZSM closed-loop system detects a massive surge in connection requests at the edge.
- **Autonomous Decisioning:** The AI-orchestrator recognizes that the high-priority **ITS Slice** requires additional capacity to maintain sub-5ms latency.
- **Dynamic Slicing:** The system borrows spectrum and compute resources from lower-priority slices (e.g., public internet or social media) and **re-allocates them to the safety-critical transportation lane.**
- **Self-Healing:** Once the traffic clears, the resources are automatically returned to their original state.

ZSM allows the network to observe congestion, infer the service impact, and adapt resources automatically to protect the safety-critical SLA.

Numerical Exercise: The Impact of Latency on Safety

To understand why 6G is the essential "nervous system" for autonomous vehicles, we must translate network latency (time) into physical displacement (distance).

1. The Baseline Variables

- **Vehicle Speed (v):** 120 km/h $\approx$ 33.3 m/s
- **The Goal:** Minimize the delay before emergency braking starts.

2. The Calculation ($d = v \times L$)

We compare the distance a vehicle travels before the "Brake" command is processed at two different latency tiers.

Note: this is only the *reaction distance* caused by network/processing latency. The vehicle's physical braking distance starts after this and must be added separately.

Numerical Exercise: The Impact of Latency on Safety

Scenario A: Standard Cloud Computing

- **Latency (L):** 150 ms
- **Reaction Distance:** $33.3 \text{ m/s} \times 0.150 \text{ s} = 5 \text{ m}$
- The vehicle travels more than a full car length before even beginning to slow down. In high-density traffic, this is often the distance between safety and a high-speed collision.

Scenario B: 6G Edge Intelligence

- **Latency (L):** 2 ms
- **Reaction Distance:** $33.3 \text{ m/s} \times 0.002 \text{ s} = 0.066 \text{ m} = 6.6 \text{ cm}$
- The reaction is near-instantaneous. The displacement is roughly the length of a smartphone, allowing the control system to start reacting almost immediately, while the final stopping distance remains governed by braking physics.

By reducing the reaction distance from meters to centimeters, 6G transforms a "best effort" network into a "safety-of-life" infrastructure.

Use Case: Tele-Operated Driving (ToD)

6G enables **Tele-Operated Driving (ToD)**, ensuring human judgment is accessible for complex "edge cases"—like non-standard construction zones or extreme weather—where AI confidence is low.

When an AI encounters an uncertain environment, it requests human assistance. A remote operator at a central station assumes control of the vehicle's "Digital Cockpit" in real-time.

Key Technical Requirements

- **Tactile Internet (Haptic Feedback):** 6G enables bidirectional haptic flow with latency (target latency in the few-ms range), allowing operators to feel steering torque and road friction as if physically present.
- **Ultra-Stable 8K Uplink:** Vehicles transmit simultaneous high-definition feeds. This requires the **eMBB slice** to maintain stable, multi-gigabit throughput to eliminate visual lag.
- **High-Security Isolation:** As highly sensitive streams, ToD sessions utilize dedicated **security tunnels** and certificate-based authentication to prevent unauthorized control commands.



Q&A - Technical Integration

"How does the network handle handover between two MEC nodes without dropping the multi-view teleoperation video/control stream?"

1. SDN-Based Bicasting

To prevent interruptions, the **SDN** controller initiates bicasting, duplicating the VR stream to both source and target MEC nodes. This ensures data is pre-positioned at the new cell site before the vehicle crosses the boundary.

2. State & Context Transfer (NFV)

NFV and **MEC** orchestration transfer or replicate session state, including control context, stream buffers, and security credentials, before the radio link changes.

3. Predictive Resource Allocation

The **ZSM** loop uses GPS trajectory and velocity to "pre-warm" the target MEC node. By predicting movement, the system spins up compute instances and reserves radio resource blocks seconds before the actual handover.

Curriculum-Specific Deep Dives and Consolidation



Mapping the ITS Solution to Student Curricula



Computing Systems and Networks

Building the 6G backbone. Responsible for **SDN/NFV** orchestration, network slicing, and achieving sub-5ms latency through **MEC** deployment.



Multimedia Systems

Enabling the immersive interface. Focuses on **8K VR** compression, **Tactile Internet** haptics, and processing high-fidelity LiDAR/Camera sensor fusion data.



Advanced Computing

Developing the autonomous brain. Implements **Federated Learning** for privacy-preserving AI and high-speed local inference for obstacle avoidance.



Big Data and Distributed Systems

Architecting global intelligence. Manages the large-scale traffic analytics and global model-weight aggregation.

Deep Dive: Computing Systems and Networks

This pathway provides the physical and logical connectivity bridging thousands of autonomous vehicles to the central cloud.

1. Network Slicing & Quality of Service (QoS)

Engineers use 3GPP Network Slicing to create service-specific logical networks. This prioritizes safety-critical "braking" commands over non-essential traffic, ensuring dedicated bandwidth and reliability.

2. Virtualization at the Edge

Through Virtual Network Functions (VNFs) and software-defined networking, intelligence is pushed to the network "Edge." Executing workloads on MEC (Multi-access Edge Computing) nodes minimizes data travel distance, enabling the sub-10ms latency essential for V2X communication.

3. Zero Trust Security Architecture

To mitigate cyber risks, this path implements a "Zero Trust" model where no device is trusted by default. All data packets are encrypted, and every vehicle or sensor requires continuous authentication via hardware security modules (HSMs).

Deep Dive: Multimedia Systems

Multimedia Systems bridge high-speed 6G connectivity with human and AI comprehension, focusing on visualization and compression for real-time situational awareness.

1. Point Cloud Compression (PCC)

Autonomous vehicles generate massive 3D LiDAR data. Multimedia engineers develop compression standards (e.g., V-PCC, G-PCC) to transmit these "point clouds" over 6G without sacrificing the spatial and semantic detail required for detection, localization, and obstacle avoidance.

2. Augmented Reality Head-Up Displays (AR-HUD)

By transforming windshields into interactive interfaces, these systems project safety markers and hazard alerts directly onto the glass. This minimizes driver distraction and improves response times in complex environments.

3. Synchronized 3D Representations

Establishing a road-side "Digital Twin" requires consistent 3D modeling. By merging vehicle camera feeds with 3D models from Roadside Units (RSUs), these systems allow vehicles to "see" around corners through a unified environmental view.

Deep Dive: Advanced Computing

The Advanced Computing pathway ensures the intelligence governing autonomous systems is both optimized and rigorously reliable, focusing on the interface between software logic and physical safety.

1. Cyber-Physical Logic

Advanced Computing engineers design the controllers that convert high-level AI decisions into deterministic physical actions. This includes precise control of braking, steering, and acceleration based on real-time sensory data and 6G network inputs.

2. Formal Verification for Safety

For autonomous systems, traditional testing is insufficient. Advanced computing applies mathematical proofs and formal verification to software architectures, providing a guarantee that the system will behave predictably in safety-critical edge cases.

3. Real-Time Multi-Agent Optimization

Managing a fleet of thousands of vehicles requires solving massive optimization problems. This involves multi-agent path-finding and collision avoidance algorithms that must run in milliseconds to maintain smooth traffic flow and safety.

Deep Dive: Big Data and Distributed Systems

The Big Data and Distributed Systems pathway is responsible for the massive orchestration required to manage the constant, high-velocity flow of information across the global ITS network.

1. High-Velocity Stream Processing

Autonomous vehicles and RSUs generate a deluge of telemetry. Distributed systems engineers build the ingestion engines that analyze millions of sensor updates per second to provide a real-time view of traffic health.

2. System Resilience & High Availability

For a safety-critical system like traffic management, downtime is not an option. This path focuses on fault-tolerant architectures and replication strategies that ensure the network remains fully operational even if a regional cloud node or data center goes offline.

3. Data Sovereignty and Governance

As data flows across different regions, it must adhere to varied privacy regulations (like GDPR). Distributed storage systems must be designed to manage "Data Sovereignty," ensuring that sensitive vehicle telemetry is stored and processed according to local laws while still contributing to global AI training.

Incident Response Workflow

Multimedia Systems

- Accident Detection: Computer vision algorithms process RSU camera feeds to detect collisions and identify vehicle IDs via sensor fusion.

Big Data & Distributed

- Impact Analysis: Data management software pipelines ingest the incident data, simulating the immediate ripple effect on city-wide traffic flow.

Computing & Networks

- Emergency Slicing: SDN controller instantly reserves a high-priority network slice for ambulance priority slice and emergency V2X signals.

Advanced Computing

- Route Adaptation: Optimization controllers recalculate safe bypass trajectories for all autonomous vehicles in the affected zone.

The Future: 6G and Semantic Communications

Shifting from transmitting "bits" to transmitting "meaning" to achieve extreme efficiency and environmental awareness.

1. Semantic Communications: Beyond Bits

- **Concept:** Instead of sending raw, heavy data (like high-definition video), the network sends the underlying "meaning" or "intent."
- **The "Dog on the Road" Example:**
 - **Current (5G):** Transmits raw 4K video stream → High bandwidth usage.
 - **Future (6G):** Transmits a compact **task-oriented representation**: "*There is a dog on the road at coordinates X,Y*" → **Can strongly reduce transmitted data when task accuracy is preserved.**
- **Impact:** Enables real-time coordination for millions of autonomous devices without saturating the spectrum.

2. Joint Sensing and Communication (ISAC)

- **Definition:** Integrating radar-like sensing capabilities directly into the communication hardware.
- **Capability:** The 6G signal itself acts as a sensor to detect:
 - Pedestrians and cyclists.
 - Road debris or obstacles.
 - Weather conditions and traffic density.
- **Benefit:** Provides "eyes" to the network, allowing vehicles to "see" around corners using reflected radio waves from other base stations.

Ethical and Regulatory Considerations

As the ITS solution transitions from technical proof-of-concept to real-world infrastructure, we must address the socio-legal challenges inherent in autonomous systems.

1. The Trolley Problem: Programmed Decision-Making

- **The Dilemma:** How do we program the AI for unavoidable accidents where every choice carries a risk of harm?
- **The Goal:** Establishing global ethical "guardrails" and standardized prioritization logic that aligns with societal values rather than proprietary corporate interests.

2. Liability and Legal Responsibility

- **The Challenge:** In a decentralized 6G environment, the failure point could be anywhere (e.g., a hardware sensor, an SDN controller, or a specific network slice).
- **Key Question:** If the network fails and a car crashes, does the liability rest with the vehicle manufacturer, the network infrastructure provider, or the human remote operator?
- **Requirement:** Clear regulatory frameworks for "Liability-as-a-Service" and immutable network logging for forensic audits.

3. Transparency: Explainable AI (XAI)

- **The Need:** Autonomous systems often operate as "black boxes." In safety-critical transportation, this is unacceptable.
- **The Solution:** Implementing **Explainable AI (XAI)**. This ensures that every high-stakes decision made by the vehicle's "brain" is transparent, traceable, and auditable by human regulators.

Industry Standards and Frameworks

The success of high-speed ITS depends on the universal adoption of standardized communication protocols across all vehicle brands and regional infrastructures.

1. 3GPP Releases 18 and 19

- **The Bridge toward 6G:** These releases define **5G-Advanced**, introducing significant enhancements for Sidelink communication (direct vehicle-to-vehicle).
- **Key Focus:** AI/ML-based air interface design, improved power efficiency, and the "Internet of Vehicles" (IoV) support.

2. The Evolution of Radio Standards

- **Legacy: IEEE 802.11p (DSRC):** Based on Wi-Fi technology; provided early V2X capabilities but suffered from range and interference issues in dense urban areas.
- **Modern: C-V2X (Cellular V2X):** Backed by the global cellular ecosystem. It offers superior range, non-line-of-sight (NLOS) performance, and a clear evolution path towards 6G NR-V2X.

3. 5G Automotive Association (5GAA)

- **The Mission:** A global cross-industry association bridging the gap between the automotive and telecommunications sectors.
- **Role:** Driving the implementation of C-V2X technology and defining the functional requirements for safety-critical services like "Remote Driving" and "Cooperative Perception."

The Global Job Market in ITS

Intelligent Transportation Systems (ITS) act as a massive economic driver, creating specialized employment across three converging sectors.

1. Key Industry Players

The ITS "War for Talent" unites traditionally separate industries:

- **Automotive OEMs: Tesla, BMW, and Mercedes-Benz** are evolving into software-centric companies focused on autonomous stacks.
- **Telco Operators: Verizon, TIM, and Vodafone** architect the 6G backbone and edge infrastructure.
- **Tech Giants: Google (Waymo), Apple, and NVIDIA** provide essential AI, cloud, and hardware-accelerated computing.

2. Specialized Roles for the Future

Graduates are positioned for high-demand, interdisciplinary roles:

- **V2X Systems Engineer:** Designing communication protocols between vehicles and infrastructure.
- **Edge Cloud Architect:** Managing MEC nodes to ensure sub-10ms latency.
- **Multimedia Vision Specialist:** Processing LiDAR and camera data for situational awareness.
- **Distributed AI Architect:** Building federated learning systems for privacy-preserving global intelligence.

Summary: The Modular Network Applications Course Recap

This slide provides a high-level summary of the course progression, highlighting how each module built toward the final ITS/autonomous-mobility case study.

1. Foundation: Enabling Technologies

- **Topic:** SDN, NFV, and Cloud Infrastructure.
- **Key Concept:** Moving from rigid hardware to a programmable, software-defined environment. This is the "Base Layer" that makes all 6G services possible.

2. Intelligence: Advanced Network Application Paradigms

- **Topic:** MEC, Slicing, Multimedia Networking, and Distributed ML.
- **Key Concept:** Building the "Intelligence Layer." Using Multi-access Edge Computing to reduce latency and Network Slicing to guarantee performance for varied services (from immersive video streams to braking signals).

3. Efficiency: Automated Management

- **Topic:** Zero-Touch Service Management (ZSM) and AI-driven Orchestration.
- **Key Concept:** Removing the human bottleneck. We explored how Closed-Loop systems automate the deployment, scaling, and repair of network functions in real-time.

4. Outcome: The Future of Mobility

- **Topic:** Autonomous ITS Ecosystem.
- **Key Concept:** The practical application. We combined all previous concepts to solve the challenges of ultra-high-speed autonomous driving, remote vehicle operation, and city-wide traffic synchronization.

The Paradigm Shift of 6G

As we conclude this course, we move away from viewing the network as a simple "pipe" for data and begin to see it as a programmable, intelligent entity.

1. The Core Pillar: Radical Integration

The defining characteristic of 6G is the seamless fusion of three formerly distinct fields:

- **Computing:** Ubiquitous processing from the core cloud to the extreme edge.
- **Communications:** Ultra-reliable, low-latency connectivity as the foundation.
- **AI (Intelligence):** Deep learning and automated orchestration embedded in every layer of the protocol stack.

2. The Network as a "Distributed Computer"

- In the 6G era, the boundaries between the "Cloud" and the "Network" dissolve.
- The infrastructure itself becomes a massive, distributed computing platform capable of running high-stakes autonomous logic, immersive multimedia processing, and real-time city-wide analytics.
- The network is no longer just *connecting* computers; it *is* the computer.



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